

Macroinvertebrate Ecology

Comparing diversity and abundance between modified and undisturbed coastal habitats



Group members

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Eww... why macroinvertebrates?



Terrestrial
bioindicator ¹



High abundance
and species
richness



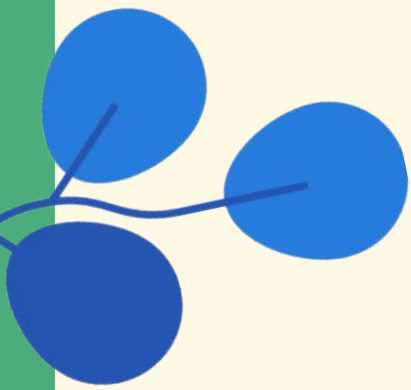
Sensitive to
changes in the
environment, e.g.
pollution ²



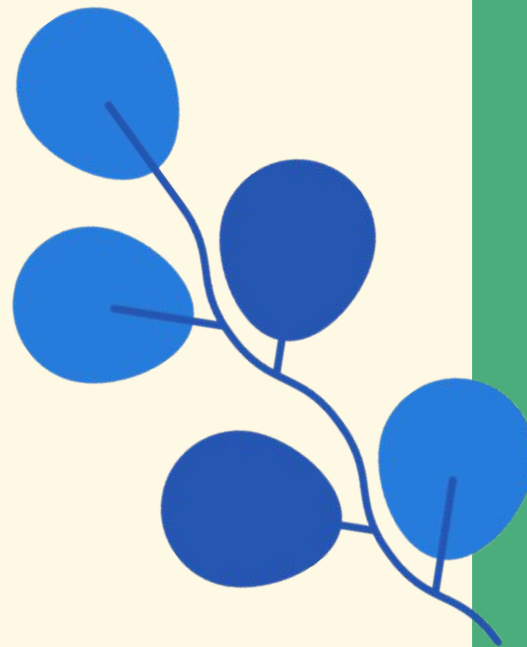
Reflect short/long
term
environmental
change



Botany Bay weevil
(*Chrysolopus
spectabilis*)



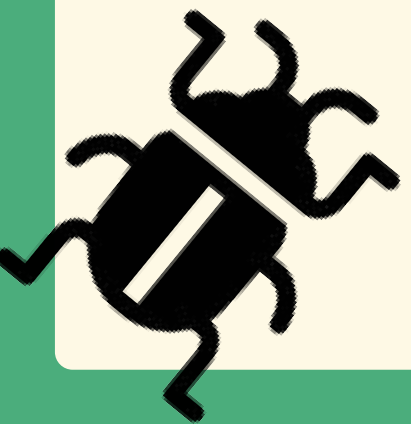
Research question



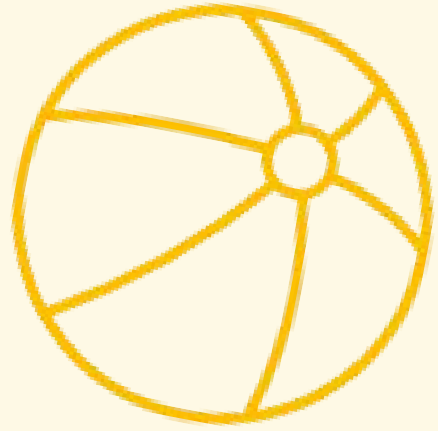
- ✓ *How does macroinvertebrate diversity and abundance differ between modified and undisturbed dune habitats?*

Aims

- ✓ Characterise macroinvertebrate Orders + richness across sites
- ✓ Characterise ant morpho-species across sites
- ✓ Compare body size between sites



Our sites

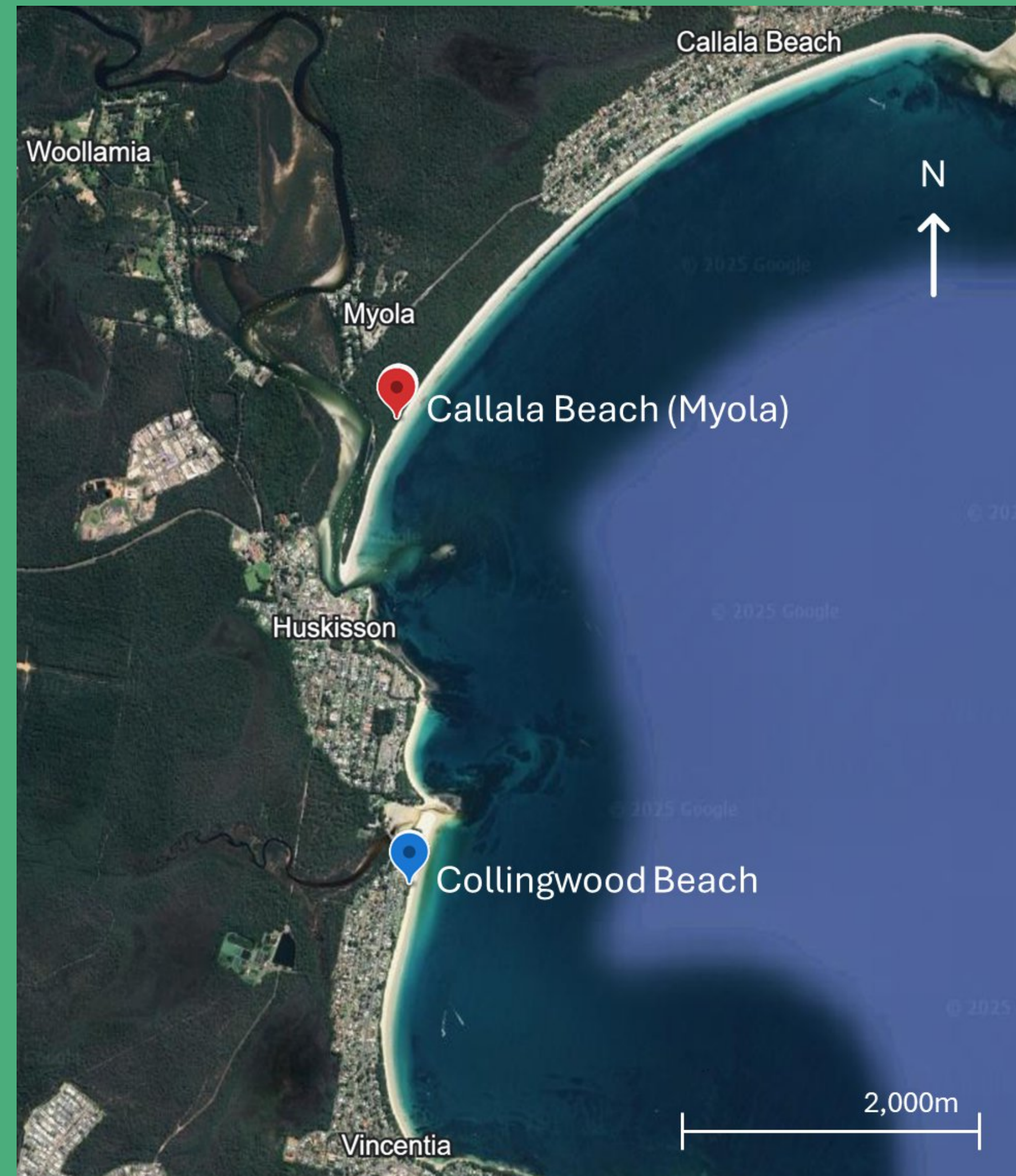


Collingwood Beach

Modified 'revegetated'
urban sand dunes

Callala Beach (Myola)

Undisturbed 'natural'
vegetated sand dunes



Modified vs Undisturbed



Collingwood Beach

- Highly urbanised on beach line
- Foredune clearing in early 1970s for urban development³
- History of unauthorised vegetation removal
- Limited canopy height, shrub dominance
- Council revegetation began in 2018³



Callala Beach

- Lower housing density; further from beach
- Relatively undisturbed; no recent clearing for urban development⁴



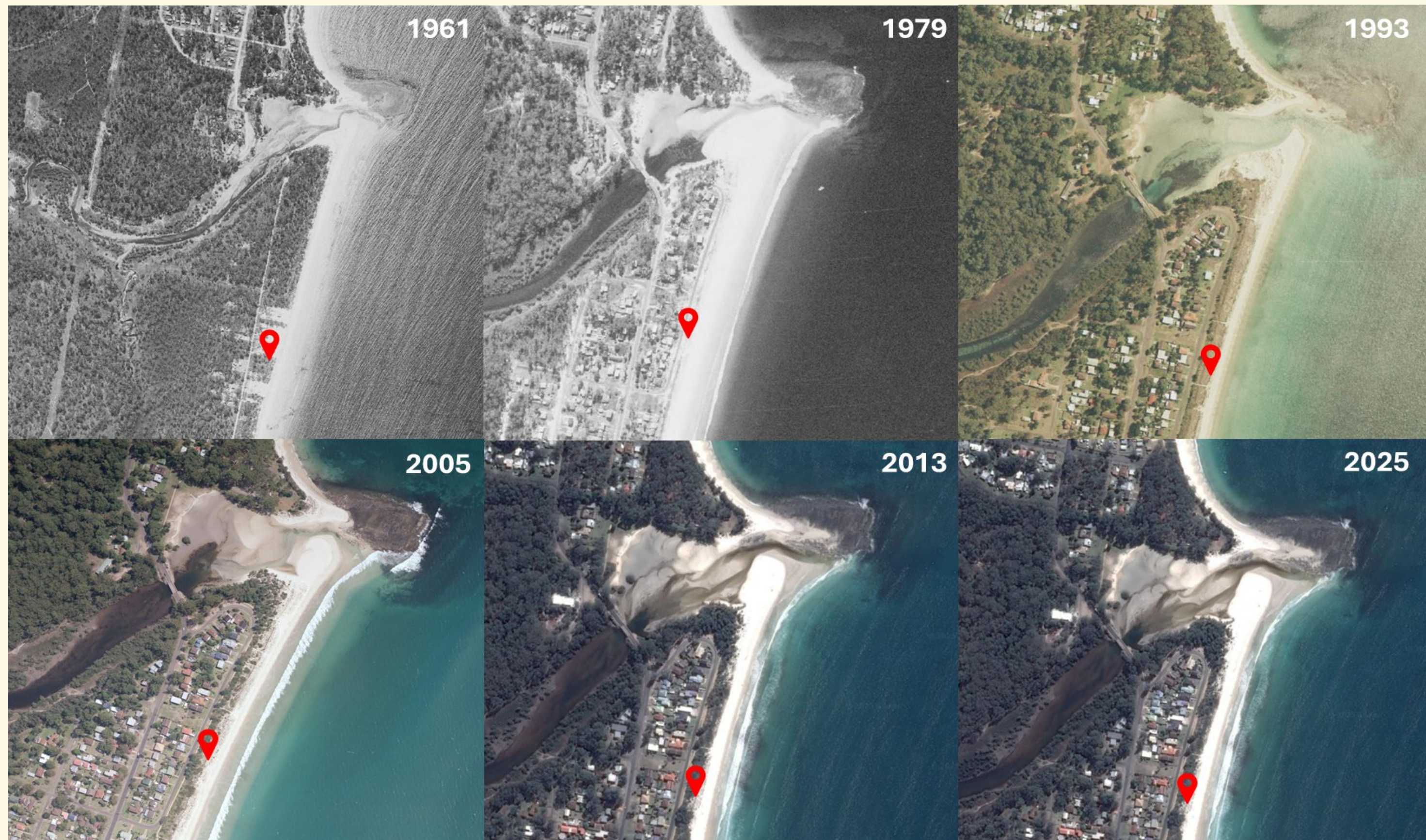


Figure 1: Historical imagery depicting foredune disturbance at Collingwood Beach (North), with markers denoting the sampling site.



Figure 2: Historical imagery depicting disturbance at Callala Beach South (Myola), with markers for the sampling site.

Methods

- ✓ Pitfall trapping
- ✓ Observation surveys
- ✓ Species ID



Observation surveys

- Multiple opportunistic methods to record macroinvertebrates:
 - A. Visual observations
 - B. Jar capture
 - C. Net capture (butterfly net)
- Surveys completed in two 3 x 3 m plots; 15 minutes per plot
- Organisms identified using Field Guides and iNaturalist



A



B



C

Pitfall trapping protocol



Body size classification

- Visual estimates undertaken during surveys
- Classified into body size categories:

< 0.5 cm
0.5-1 cm
1-1.5 cm
1.5-2 cm
> 2 cm



Identifying ants to morphospecies

- Assessed under dissecting microscope
- Individuals classified into morphospecies based on key traits:
 1. Body segmentation, shape
 2. Antennae structure, segments
 3. Mandible shape
 4. Spines, hairs, surface textures
 5. Node structure on petiole



Figure 4. Laboratory microscope and ant viewed under magnification.



Results



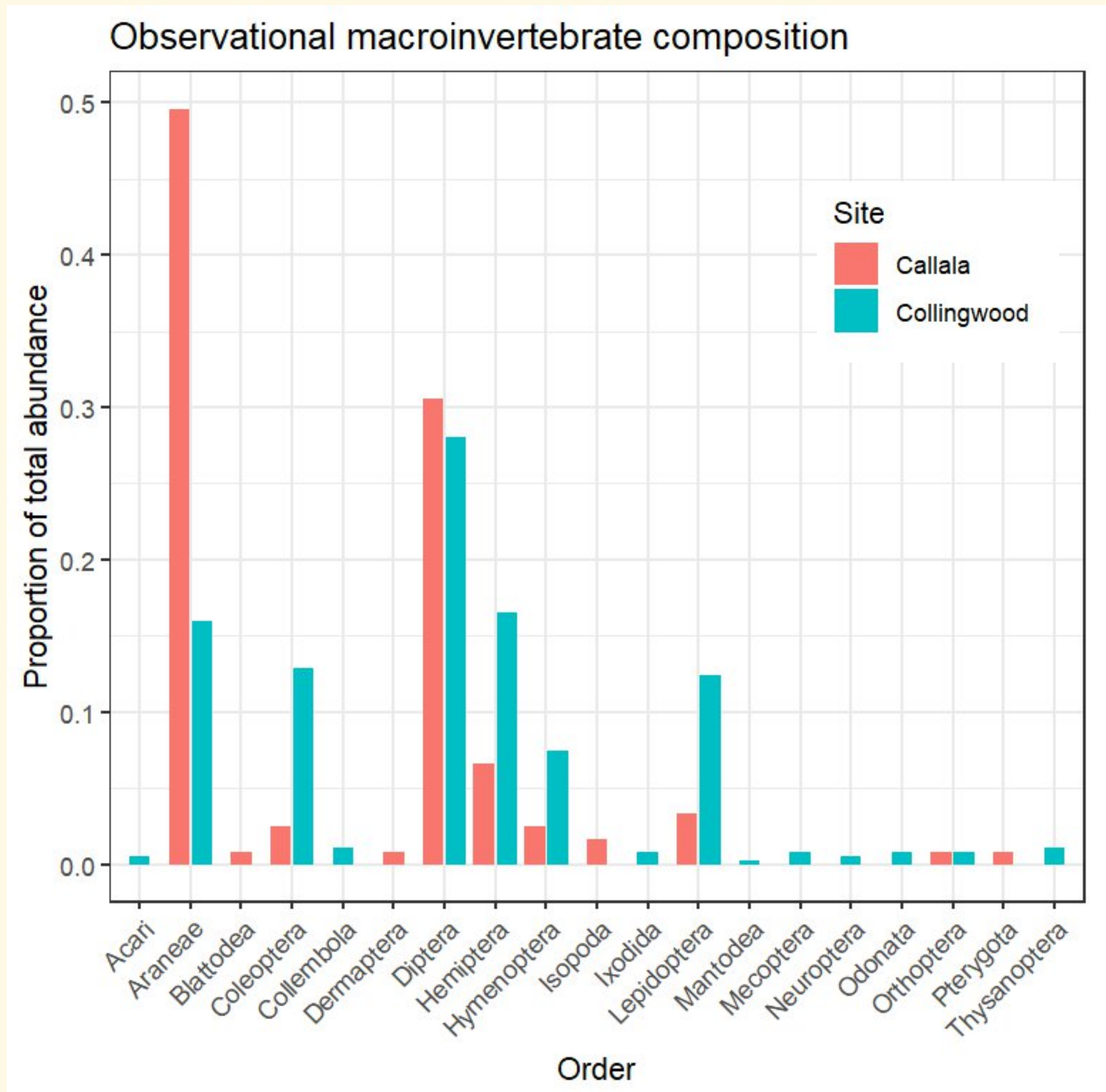
*C.
spectabilis ..
.again*

Table 1. Summary of macroinvertebrate counts at Collingwood and Callala Beach

	Collingwood	Callala
Total Count	814	687
Total observation count (with ants)	550	356
Total observation count (no ants)	369	121
Total pitfall count (with ants)	264	331
Total pitfall count (no ants)	41	119
# Orders present (pitfall)	10	13
# Orders present (observation)	15	11

Observation results

Figure 4. Proportion of macroinvertebrate Orders from observation surveys at Collingwood Beach and Callala Beach, with ants excluded from *Hymenoptera*.



Shannon's Diversity for observation results



Table 2. Welch's t-test: Overall observational diversity (H)

COL	CAL	t-value	df	p-value
1.658	0.975	-4.921	7.45	0.00143

And without ants... same!



COL	CAL	t-value	df	p-value
1.729	0.992	-6.499	10.78	0.0000488



Greengrocer cicada
(*Cyclochila australasiae*)

Pitfall results

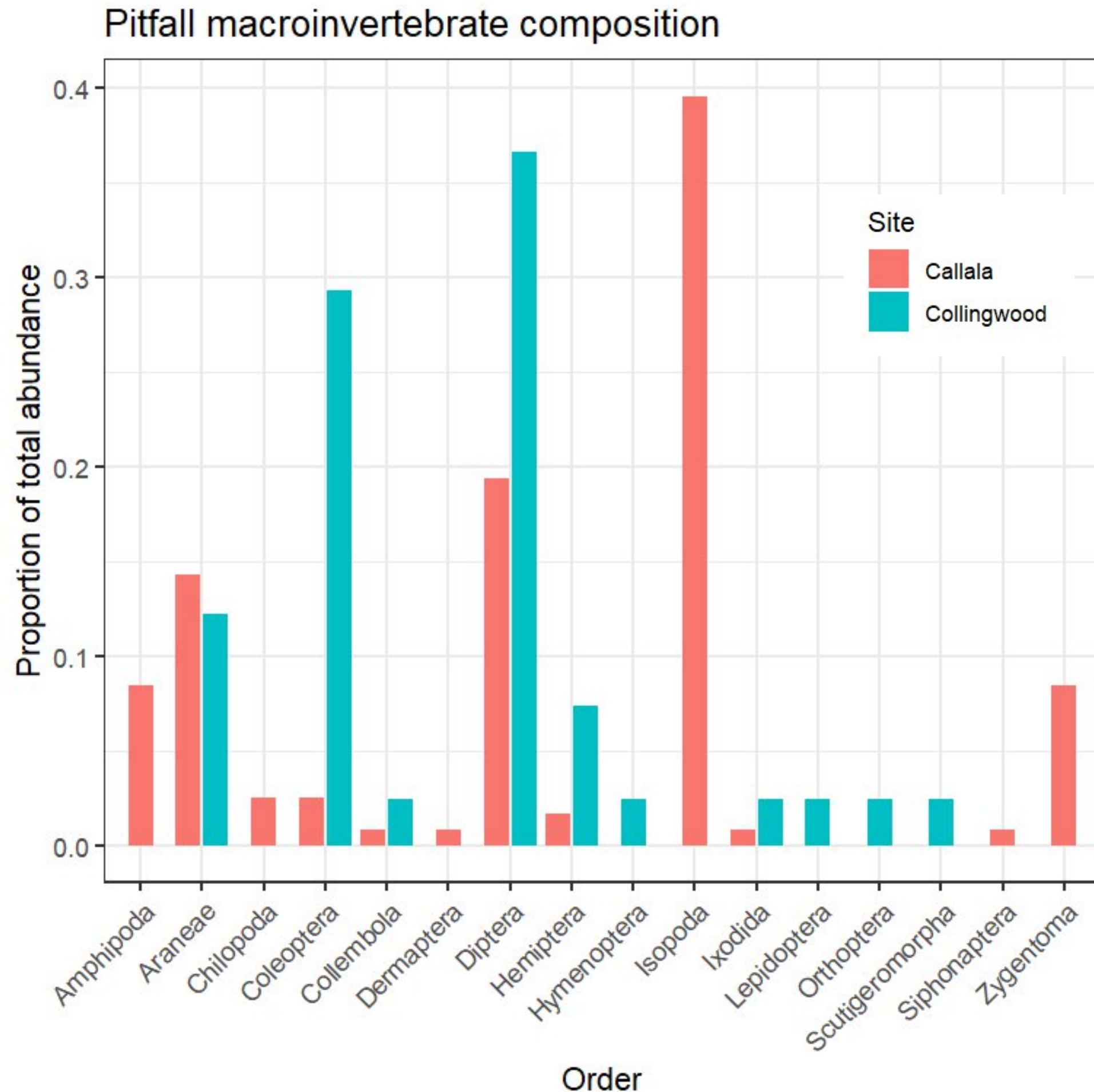
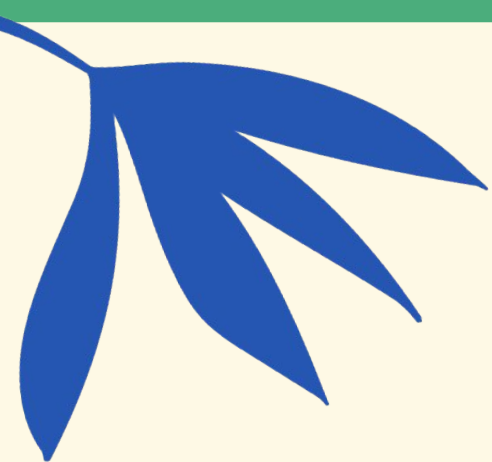


Figure 5. Proportion of macroinvertebrate Orders from pitfall traps at Collingwood and Callala Beach, with ants excluded from *Hymenoptera*.





Site differences in pitfalls

Table 3. Shannon's Diversity Index (H) Day and night traps.

	Day	Night	df	p-value
COL	0.118	0.519	26.06	0.0021
CAL	0.558	0.682	16.85	0.4412

Table 4. Welch's t-test: Overall pitfall diversity (ants *included*).

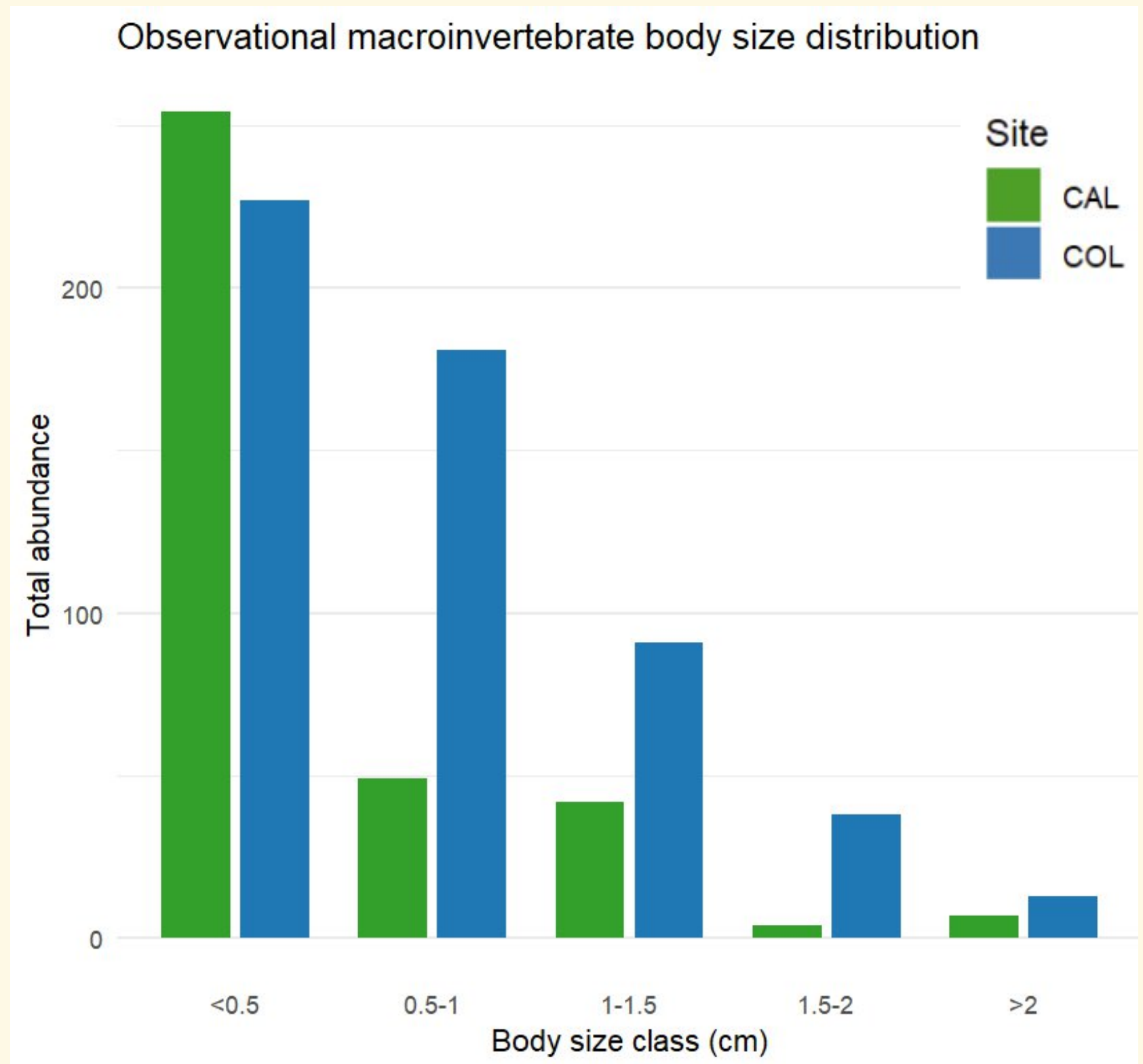
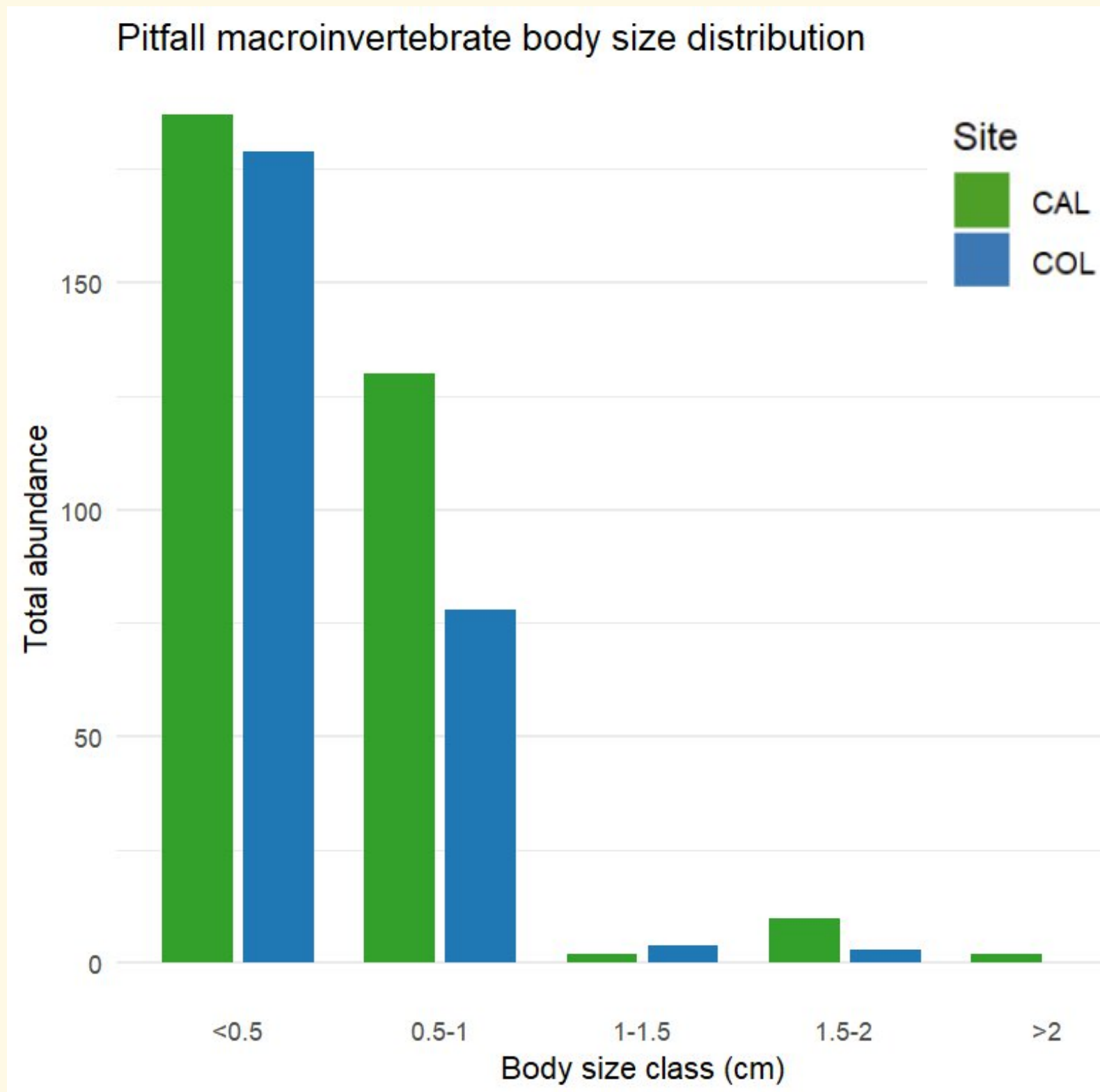
COL	CAL	t-value	df	p-value
0.413	0.862	3.87	30.319	0.000537

Table 5. Welch's t-test: Overall pitfall diversity (ants *excluded*).

COL	CAL	t-value	df	p-value
0.466	0.712	1.57	23.17	0.1301



Differences in body size



Ant morphospecies

Table 6. Presence of distinct ant morphospecies at each site.

Scientific Name	CAL	COL
<i>Alates spp</i>	✓	✗
<i>Camponotus spp</i>	✗	✓
<i>Linepithema spp</i>	✓	✗
<i>Ochetellus spp</i>	✓	✓
<i>Paratrechina spp</i>	✓	✓
<i>Pheidole spp</i>	✓	✓
<i>Polyrhachis spp</i>	✓	✗
<i>Rhytidoponera spp</i>	✓	✓
<i>Solenopsis spp</i>	✓	✓

Table 7. Shannon’s Diversity Index (H) and evenness for ant morphospecies across sites.

	CAL	COL
Shannon H	1.26	1.29
Evenness	0.61	0.72



Rhytidoponera spp



Pheidole spp



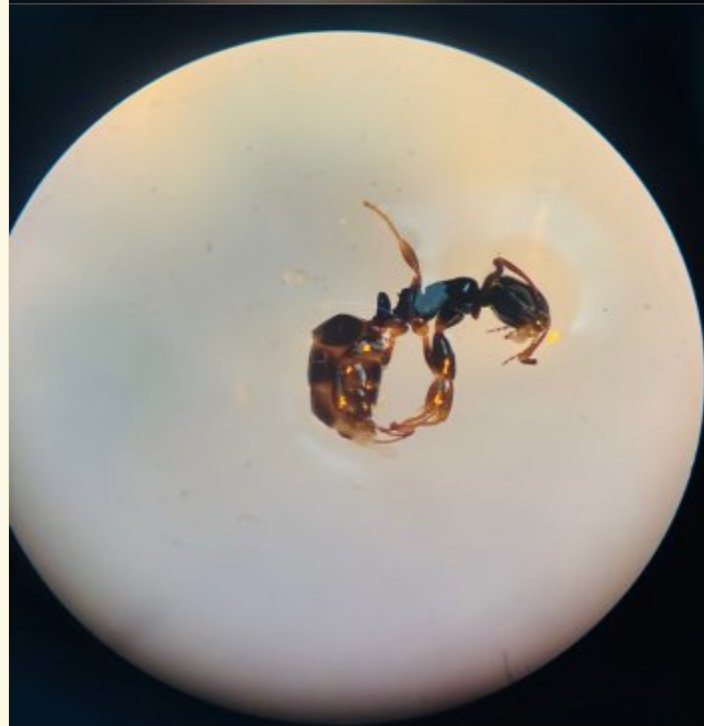
Camponotus spp



Ochetellus spp



Paratrechina spp



Linepithema spp



Solenopsis spp



Polyrhachis spp

Dominant genera

Collingwood: Crazy ants
(*Paratrechina* spp; N = 133, 60.7%)

Callala: Big-headed ants
(*Pheidole* spp; N = 98, 57.65%)

Discussion

- Observation diversity: COL > CAL
 - Not necessarily more structurally/functionally diverse
 - More opportunistic species present?
- Pitfall diversity: CAL > COL
 - Driven by ant dominance
- Morphospecies diversity: CAL > COL
 - Not necessarily positive; some non-native, invasive
- Body size: more large invertebrates in observations
 - Larger size at Collingwood



Limitations

- Site replication (only one pair)
- Temporal replication (one sampling period)
- Proximity of plots to pitfall traps
- Short survey window (15 min)
- Observer bias: searching effort, ID skill, detection
- Taxonomic resolution: ID to Order, not species
- No consideration of environmental variables



Conclusions

- Different sampling methods detected different communities
 - Effective approach for distinguishing modified vs unmodified dune habitats
- Higher diversity in disturbed site may reflect invasive or opportunistic taxa
- More reference sites needed to determine observed patterns at both sites



Acknowledgements

- Rachael for keeping us 'tick'ing!
- Botany Bay weevils for keeping everyone happy
- Field groups for collecting our data
- Simon for coordinating the trip
- Max, Jess, Adam and Kerrie for their assistance in the field
- And a big thanks Nico for being our personal Uber to sites :)



Last weevil pic... we

swear

Any questions?



Leafhopper
(*Ledromorpha planirostris*)

References

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